1 Project Overview

1.1 Project Details

Project title: Loan Approval Prediction Analysis
Role: Data Analyst
Primary stakeholder: Head of Credit Risk

1.2 Executive Summary

2 Project Background

Lending institutions receive a high volume of loan applications, each requiring an assessment of the applicant's likelihood of repayment before a decision is made. This process traditionally relies on manual underwriting where credit officers review financial and credit history details case by case. As application volume grows, this manual review process becomes a bottleneck, and decisions can vary in consistency depending on the reviewer.

The Head of Credit Risk wants to understand whether the bank's historical approval decisions follow predictable patterns based on applicant characteristics such as credit score, income, and asset holdings. Building a model that reflects these patterns would help the team identify which applications are strong candidates for fast-track approval, which carry clear rejection signals, and which genuinely need manual review because the decision isn't obvious from the data alone.

3 Business Problem

Manual loan application review is time-consuming and can produce inconsistent outcomes when different reviewers assess similar applicant profiles. Without a data-driven reference point, the Head of Credit Risk has limited visibility into which applicant characteristics most strongly influence approval outcomes, or which applications are straightforward enough for fast-track review versus those that require closer assessment.

This project addresses that gap by using historical applicant and approval data to build a classification model that estimates loan approval likelihood, providing the credit risk team with a data-backed reference point to support faster, more consistent application review.

4 Objective

This project aims to give the Head of Credit Risk a data-driven view of loan approval patterns, specifically to:

- Identify which applicant characteristics (credit score, income, loan amount, assets, employment status) most strongly influence approval outcomes
- Build a classification model that predicts the likelihood of loan approval based on applicant data
- Provide a reference tool that can help flag applications suited for fast-track processing versus those needing closer manual review

Professional development objective:

This project extends beyond descriptive analytics into predictive modelling, demonstrating hands-on experience with train/test splitting, model training, and classification evaluation metrics such as precision, recall, and ROC-AUC.

5 Scope

5.1 In scope

- Loan applicant dataset covering demographic, employment, financial, credit, and asset-related fields, along with the historical loan approval outcome
- Data Cleaning and preparation in Python
- Exploratory analysis of approval patterns across key variables (credit score, income, assets, employment status)
- Classification model to predict loan approval likelihood, using this dataset only
- Model evaluation using standard classification metrics
- Power BI dashboard summarizing EDA findings and model results

5.2 Out of scope

- Any data outside this dataset (no external credit bureau data, macroeconomic factors, or additional applicant history)
- Model deployment or integration into any live application system
- Advanced or ensemble modelling techniques beyond a small comparison of simple classifiers (this is a learning project, not a production model)
- Bias mitigation or fairness correction techniques (flagged for awareness in Section 14, but not actively corrected for within this project's scope)

6 Data Sources

This project uses a loan applicant dataset from Kaggle containing demographic, employment, financial, credit, and asset-related information, along with the historical loan approval outcome.

Source: Loan Approval Dataset

7 Key Questions

1. Which applicant characteristics show the strongest relationship with loan approval outcomes (credit score, income, loan amount, assets, employment status, education)?
2. Does the Credit Information Bureau (India) Limited, CIBIL score alone explain most approval decisions, or do other factors meaningfully affect the outcome even after accounting for credit score?
3. How well can classification model predict loan approval status using the available applicant data?
4. Which features does the model rely on most heavily, and does this match what the exploratory analysis shows?
5. Where does the model perform poorly (false positives or false negatives), and what does that suggest about the limits of using this data alone for approval decisions?

8 Deliverables

- Raw and cleaned dataset (CSV) , stored separately
- Jupyter notebook(s) covering data cleaning, exploratory data analysis, feature preparation, model training, and evaluation
- MySQL database containing the cleaned data, used for EDA-stage querying
- Power BI dashboard covering exploratory findings (approval rate by CIBIL band, income, employment status, etc.) and model results (confusion matrix, feature importance)
- Model evaluation summary covering accuracy, precision, recall, and ROC-AUC
- Project documentation, including the README and project charter

9 Tools and methodology

Python (pandas, numpy) : For data cleaning, feature preparation, and exploratory analysis

Scikit-learn	:	For training the classification model and calculating evaluation metrics (precision, recall, ROC-AUC)
Matplotlib / Seaborn	:	For exploratory visualizations within the notebook during analysis
Jupyter Notebook	:	Environment for writing and documenting the analysis and modelling workflow
MySQL	:	For storing cleaned data and running EDA-stage queries on approval patterns
Power BI	:	For building the stakeholder-facing dashboard covering EDA findings and model results
Git/GitHub	:	For version control and publishing the project to the portfolio

10 Analysis Workflow

Data Loading and Cleaning	:	Load the raw dataset into Python, check for missing values, duplicates, and inconsistent formatting (e.g. stray whitespace in categorical fields), and validate data types
Exploratory Data Analysis	:	Examine distributions of key variables (CIBIL score, income, loan amount), and analyse approval rates across segments (credit score bands, income brackets, employment status, education)
Feature Preparation	:	Encode categorical variables (education, self-employment status) into numeric form, check for class imbalance in the target variable, and split data into training and test sets
Model Training	:	Train a baseline classification model (logistic regression), then train a simple comparison model (decision tree) to see how performance differs
Model Evaluation	:	Assess both models using accuracy, precision, recall, and ROC-AUC, then select the stronger baseline based on business relevance and balanced classification performance.
Interpretation	:	Examine feature importance to understand which variables drive the model's predictions, and compare this against what the EDA showed
Dashboard	:	Build the Power BI dashboard summarizing EDA findings and model results
Documentation	:	Write the README and publish the project repository to GitHub.

11 Project Roadmap

Day	Focus	Tasks
Day 1	Data Loading and Cleaning	Load dataset, check for missing values/duplicates, validate data types
Day 2	Exploratory Data Analysis	Analyse distributions and approval rates across key segments
Day 3	Feature Preparation and Model Training	Encode categorical variables, split data, train logistic regression and decision tree models
Day 4	Model Evaluation and Interpretation	Evaluate models using classification metrics, examine feature importance
Day 5	Dashboard Development	Build Power BI dashboard covering EDA and model results
Day 6	Documentation and Publishing	Write README, finalize charter, publish to GitHub

12 Success Criteria

This project will be considered successful if:

- The dataset is fully cleaned and validated, with no missing values or inconsistent formatting remaining
- All key questions defined in this charter can be answered through exploratory analysis
- At least one classification model is trained and evaluated using precision, recall, and ROC-AUC, not accuracy alone

- The evaluation results are reported honestly, including cases where model performance is limited or the model struggles with certain segments
- Feature importance findings align logically with what the exploratory analysis shows, or any mismatch is explained
- The Power BI dashboard clearly communicates both the EDA findings and the model's performance to a non-technical stakeholder.
- The README and charter honestly frame this as a stretch/learning project, not a production-ready predictive system

13 Assumptions and dependencies

13.1 Assumptions

- The historical loan approval decisions in the dataset reflect reasonable lending policy at the time they were made. This project models what the bank's past decisions were, not what they should have been, and does not attempt to correct or second-guess those historical outcomes
- The dataset accurately represents real or realistic loan applicant scenarios, even if anonymized or synthetically generated
- CIBIL score and other financial fields are recorded consistently and correctly at the time of application
- The dataset is complete enough (in terms of features) to build a reasonably informative model, though it may not capture every factor a real underwriter would consider

13.2 Dependencies

- Availability and stability of the Kaggle dataset link throughout the project
- Python libraries (pandas, scikit-learn, matplotlib/seaborn) functioning as expected in the existing Anaconda environment
- MySQL Workbench and Power BI Desktop being available and functioning on the local machine

14 Model Limitations and Ethical Considerations

14.1 Limitations

- This model is trained on a single historical dataset and has not been tested against real-world applicants outside this data, so its predictions may not generalize beyond the patterns present here
- The dataset may not capture every factor a real underwriter considers (such as full credit history details, employment stability over time, or macroeconomic conditions), meaning the model's view of "approval likelihood" is necessarily incomplete
- As a first classification project, model performance is expected to be a learning benchmark rather than a production-grade result, and any limitations in accuracy or reliability should be reported honestly rather than adjusted to took stronger than they are

14.2 Ethical Considerations

- Loan approval prediction is sensitive use case, since biased historical lending patterns (even if unintentional) can be learned and reproduced by a model trained on them
- This project does not include fairness auditing or bias mitigation techniques. Because these are advanced topics outside the scope of this first predictive modelling project, the limitation is explicitly acknowledged rather than ignored.
- If this model were ever considered for real-world use, it would first require a proper fairness assessment across demographic and socioeconomic groups, along with human oversight, before being used to influence actual lending decisions

15 Skills Demonstrated

15.1 Technical Skills

- Python (pandas, NumPy) for data cleaning and exploratory analysis
  - Scikit-learn for building and evaluating classification models
  - Matplotlib / seaborn for exploratory data visualization
  - MySQL for structured data storage and querying
  - Power BI for dashboard development
- Git / GitHub for version control and portfolio publishing

15.2 Analytical Skills

- Exploratory data analysis and segment-based pattern identification
- Binary classification modelling
- Model evaluation using precision, recall, and ROC-AUC
- Feature importance interpretation
- Awareness of model limitations and ethical considerations in predictive analytics